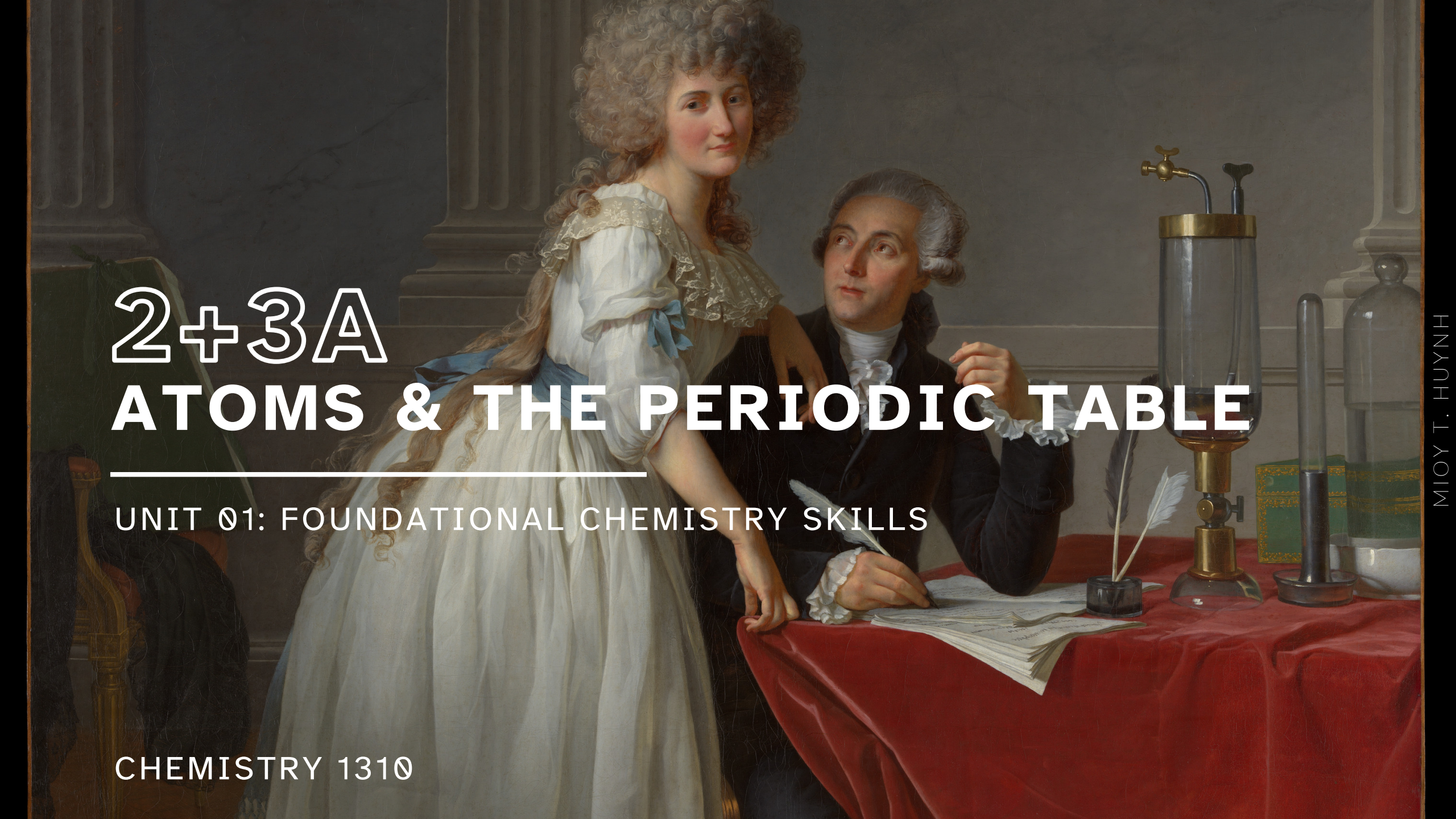


A classical painting of Antoine Lavoisier and Laplace in their laboratory. Lavoisier is seated at a table covered with a red cloth, writing on a document. Laplace stands beside him, looking at the document. On the table are various scientific instruments, including a large glass apparatus with a pump, a thermometer, and a balance scale. The background features classical columns and a dark, moody atmosphere.

# 2+3A ATOMS & THE PERIODIC TABLE

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UNIT 01: FOUNDATIONAL CHEMISTRY SKILLS

CHEMISTRY 1310



# LEARNING OBJECTIVES

## CHAPTER 2

1. Recognize the symbols and names of common elements.
2. Outline Dalton's atomic theory.
3. Apply Rutherford's nuclear model of the atom.
4. Identify the mass number and atomic number of an atom using the count of subatomic particles present and vice versa.
5. Identify isotopes of element using their atomic and mass numbers; count subatomic particles in a given isotope.
6. Calculate the atomic mass of an element from the masses and abundances of its naturally occurring isotopes.
7. Classify elements in a systematic manner based on their location on the periodic table.
8. Identify periods, groups, and sections of the periodic table.

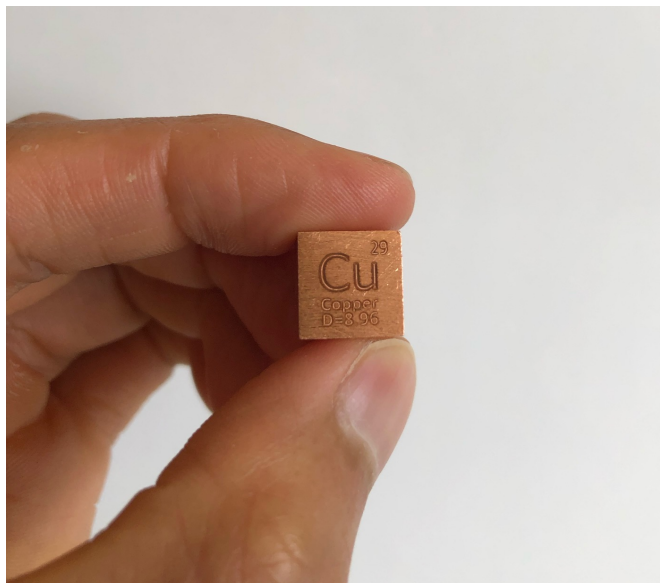
# Domains of Chemistry

In this chapter, we'll discuss atomic structure, but it's important to remember that we can't actually "see" atoms.

## MACROSCOPIC

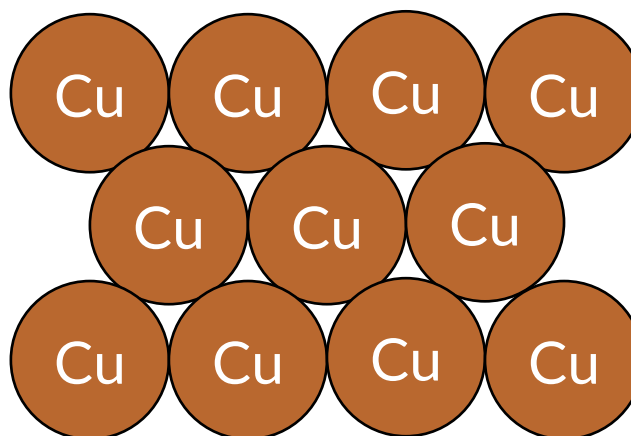
"large"

Can see with naked eye



## SYMBOLIC

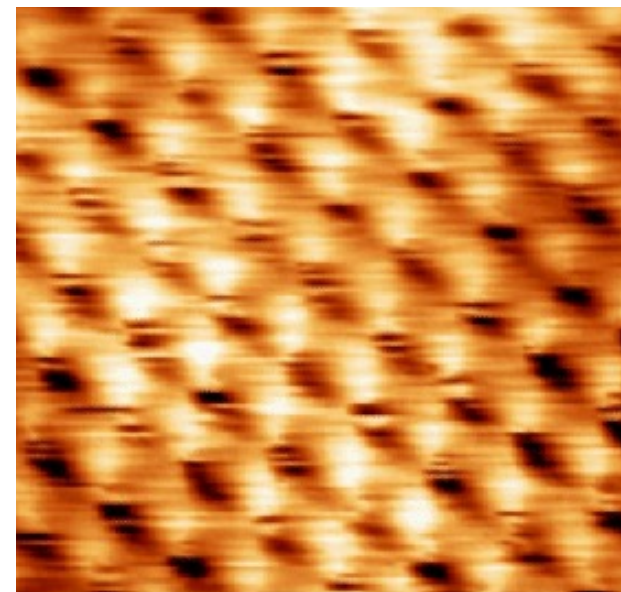
Used to portray macroscopic and microscopic domains



## MICROSCOPIC

"small"

Requires equipment to "see"



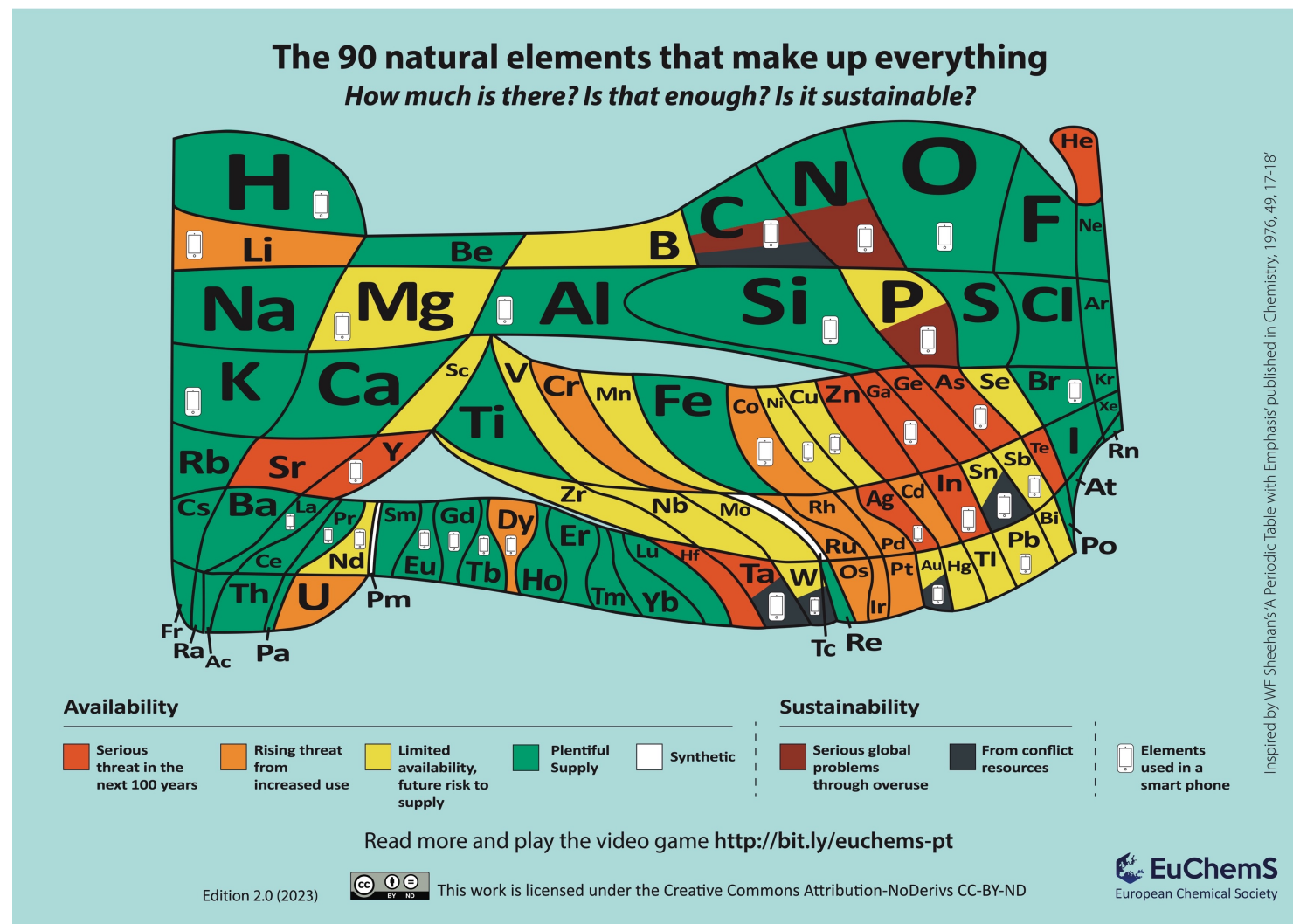
# Chemical Symbols

## 2.1

Each of the 118 known chemical elements is identified by an internationally recognized symbol consisting of either:

- one capital letter (C, H, O, F, etc.)
- one capital letter followed by a lower-case letter (Na, Fe, Cl, Ar, etc.)

There are 10 elements whose symbols are derived from the chemical name in a language other than English.

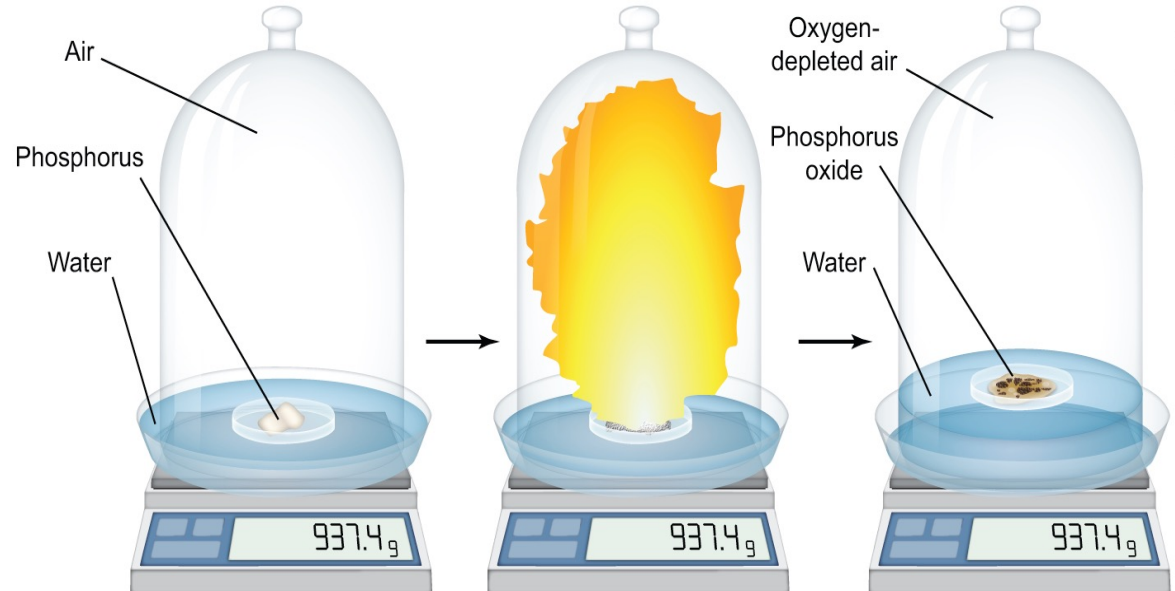


# Laws of Chemical Composition

## 2.2

The mass of the products generated during a chemical reaction is equal to the mass of reactants used up during the reaction.

This observation is the basis of the law of conservation of mass.



**FIGURE 2.1** An Experiment to Illustrate the Law of Conservation of Mass

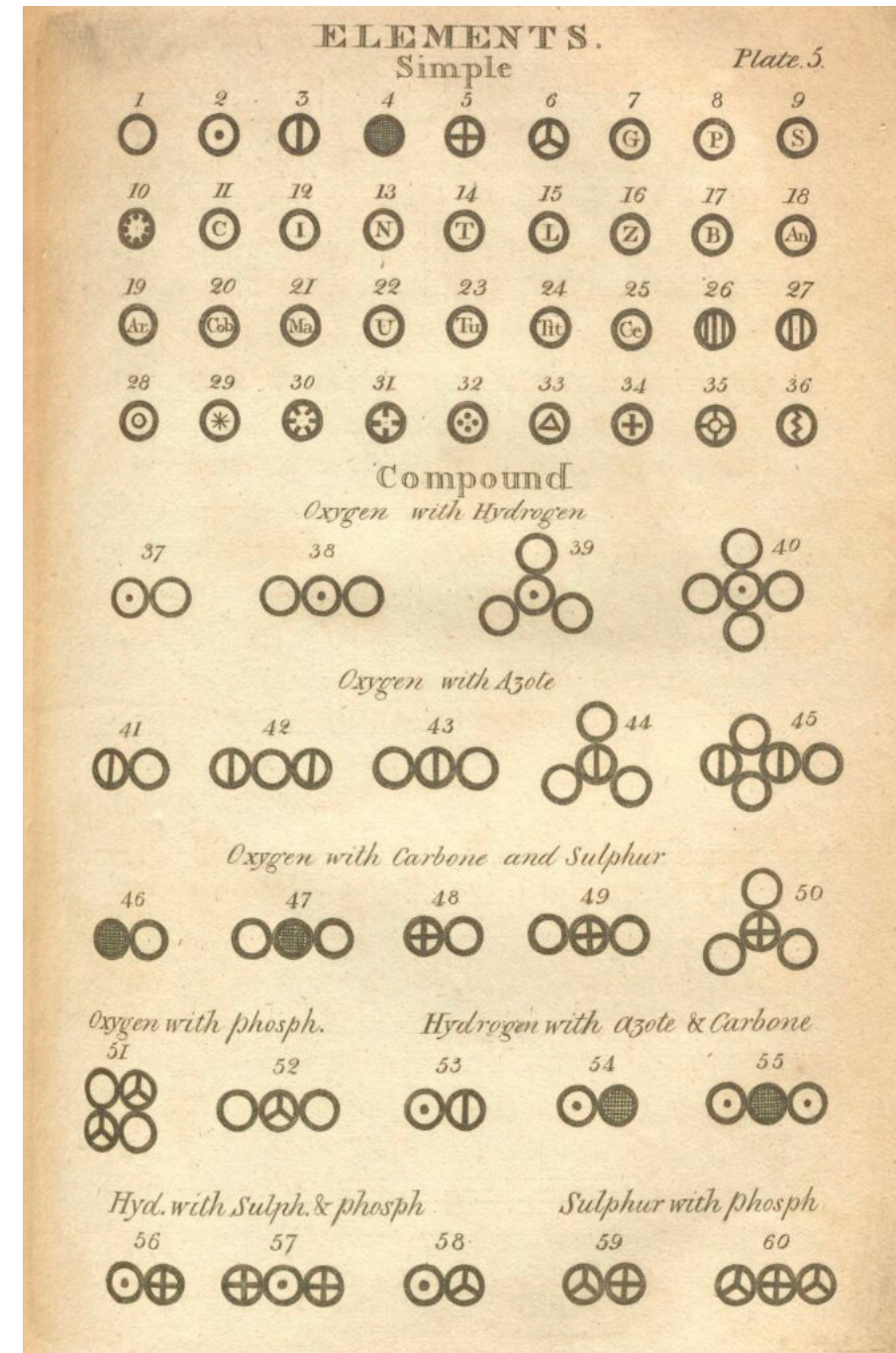


# The History of the Atom

## 2.3

John Dalton formulated his [atomic theory](#) to explain the different laws of chemical combination:

1. Matter is made up of very tiny, indivisible particles called [atoms](#).
2. Each atom of a particular element has the same mass, but the mass of an atom of one element is different from the mass of an atom of any other element.
3. Atoms combine to form what we now call [molecules](#). When they do so, they combine in small, whole-number ratios.
4. Atoms of some pairs of elements can combine with each other in different, small, whole-number ratios to form different compounds.



# The History of the Atom

## 2.3

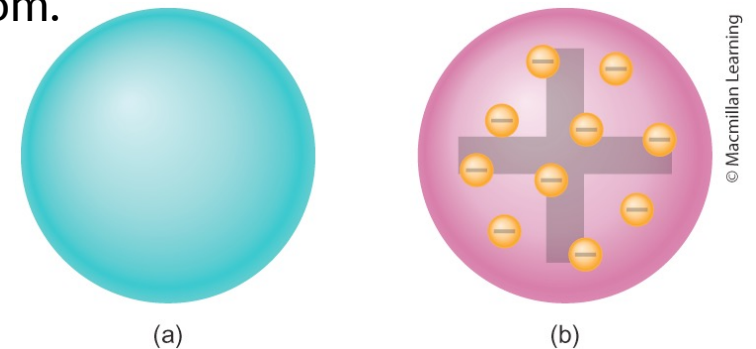
English physicist J.J. Thomson discovered the negatively charged subatomic particle now known as the electron in the late 1800s.

American physicist Robert Millikan succeeded in measuring the mass of a single electron in the early 1900s.

- Because atoms are neutral overall, the “non-electron” part of the atom must have a positive charge equal to the negative charge of electrons.
- The electron accounted for only  $1/1836$  of the mass of a hydrogen atom.
- These observations led to Thomson’s plum-pudding model of the atom.

Dalton’s early model (a) consisted of an indivisible sphere.

Thomson’s plum-pudding model (b) consisted of negatively charged electrons suspended in a larger sphere of positive charge.



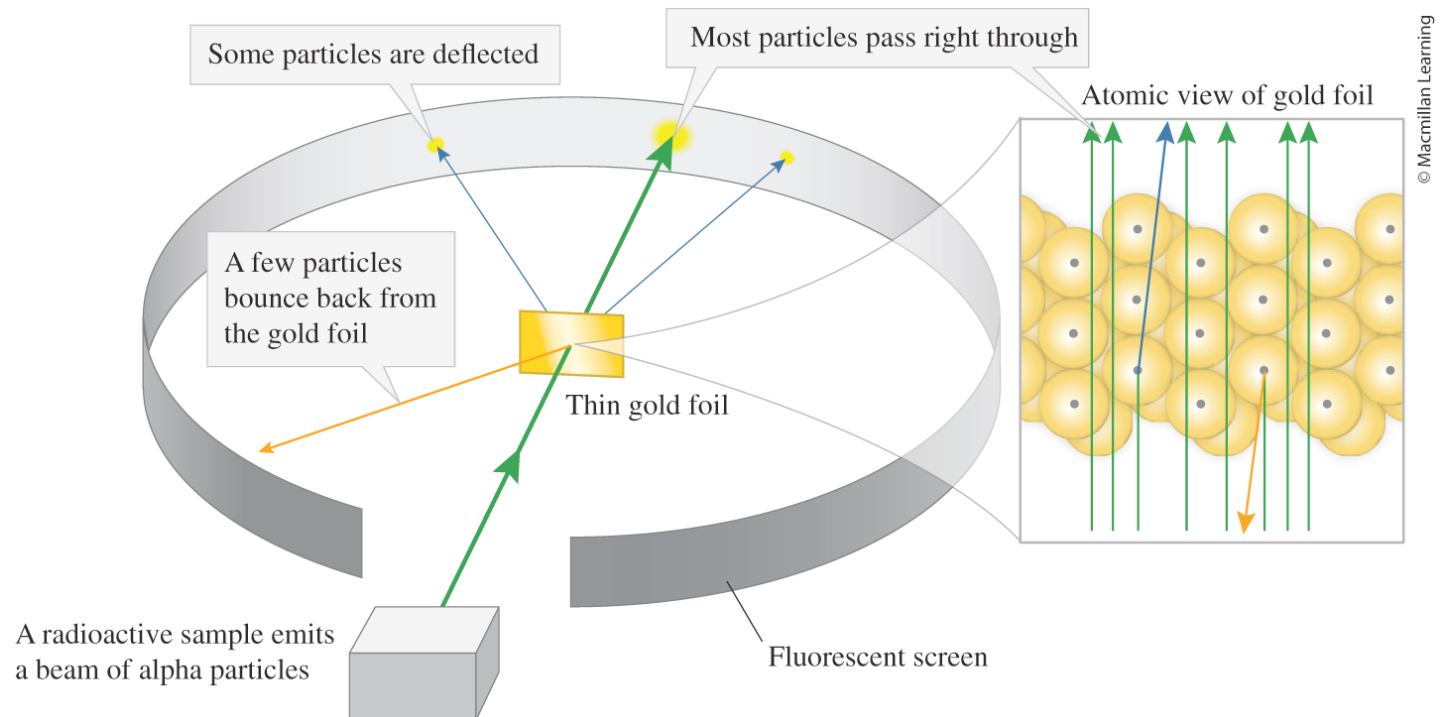
**FIGURE 2.7 Dalton and Thomson’s Early Models of the Atom**

# The History of the Atom

## 2.3

British scientist Ernest Rutherford's gold foil experiment was designed to substantiate Thomson's plum-pudding model.

Instead, it led to a new nuclear model of the atom.



**FIGURE 2.8 Rutherford's Gold Foil Experiment**

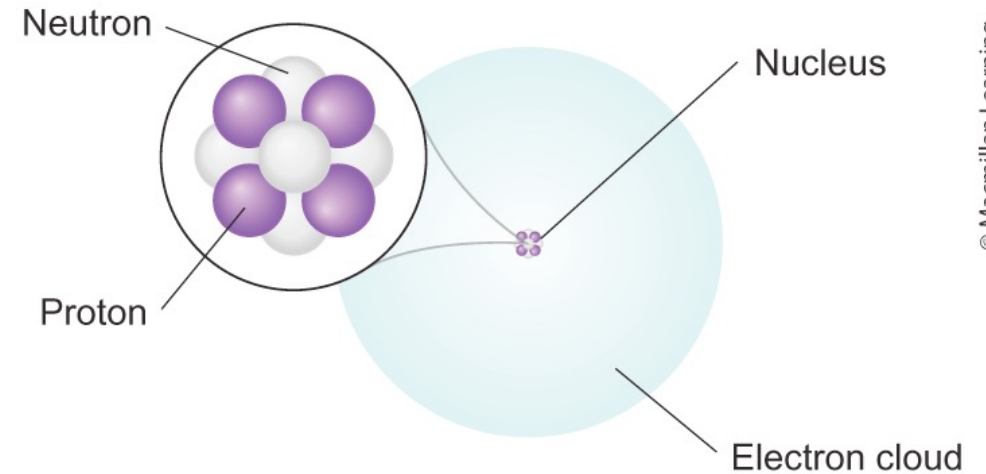


# The History of the Atom

## 2.3

Rutherford's nuclear model:

- The atom's positive charge and the majority of its mass are located in a relatively small area, the nucleus.
- The vast majority of the atom is empty space.
- Small, negatively charged electrons are spread throughout this empty space.
- The number of negatively charged electrons is equal to the number of positively charged particles (protons) inside the nucleus.



**FIGURE 2.9 Nuclear Model of the Atom**

# Subatomic Particles, Isotopes, and Ions

2.4

Protons, neutrons, and electrons are subatomic particles.

- The nucleus comprises almost all of the *mass* of an atom whereas most of the *volume* is occupied by electrons.

| Particle                   | Charge (e) | Mass<br>(atomic mass units) | Location in the atom |
|----------------------------|------------|-----------------------------|----------------------|
| Proton (p <sup>+</sup> )   | +1         | 1.0073                      | In the nucleus       |
| Neutron (n)                | 0          | 1.0087                      | In the nucleus       |
| Electron (e <sup>-</sup> ) | -1         | 0.000549                    | Outside the nucleus  |

- Note: The charges given are *relative* charges, based on the charge of the electron (denoted as *e*) as the fundamental unit of charge.

$$1\ e = -1.60 \times 10^{-19}\ \text{Coulombs (C)}$$

# Subatomic Particles, Isotopes, and Ions

2.4

Atoms are electrically neutral:

Atomic number,  $Z$ , is the number of protons in the nucleus of an atom.

- All atoms of a given element have the same number of protons.



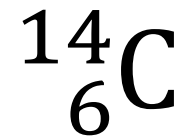
Mass number,  $A$ , is the number of protons plus the number of neutrons in an atom.



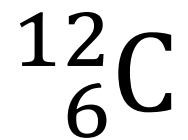
# Subatomic Particles, Isotopes, and Ions

2.4

Isotopes are atoms of the same element that have different numbers of neutrons.



vs.

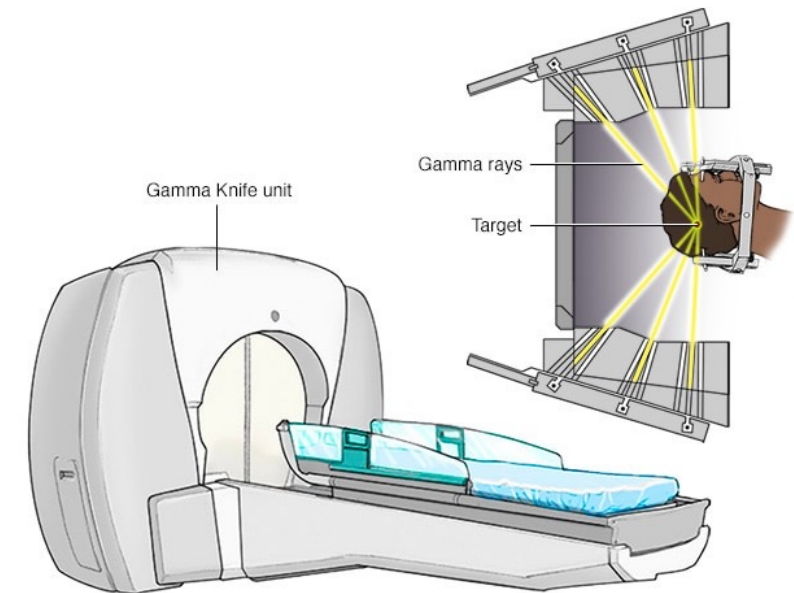


## EXAMPLE PROBLEM 1

Learning Objective(s): 2.5

An atom of  $^{210}\text{Pb}$  has how many neutrons?

How many neutrons are there in an atom of the isotope  $^{60}\text{Co}$ ?



# Subatomic Particles, Isotopes, and Ions

2.4

Ions: Species with an overall net charge because the numbers of protons and electrons are not equal.

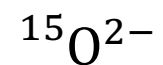
- Anions: more electrons than protons
- Cations: more protons than electrons

Ions can be either monatomic (derived from a single atom) or polyatomic (a group of atoms with an overall charge).

## IN-CLASS QUESTION 1

Learning Objective(s): 2.5

Determine the number of protons, neutrons, and electrons in the ion below.





## EXAMPLE PROBLEM 2

Learning Objective(s): 2.1, 2.4

How many protons and electrons are in the chlorine atom? In the chlorine ion ( $\text{Cl}^-$ )?

# Atomic Masses

2.5

The masses of atoms that we use today are on a relative scale.

- The mass of one isotope of one element was assigned a value, and the masses of all other atoms are measured relative to that standard.

## Average atomic mass

- The weighted average of the masses of the naturally occurring mixture of isotopes of an element, compared with one-twelfth of the mass of a single  $^{12}\text{C}$  atom.
- Found on the periodic table along with the atomic number for every element.
- *Also called: atomic mass, atomic weight, or average atomic weight.*

## EXAMPLE PROBLEM 3

Learning Objective(s): 2.6

Antimony, Sb, has two stable isotopes with the following atomic masses (in amu):

$$^{123}\text{Sb} = 122.904 \text{ amu}$$

$$^{121}\text{Sb} = 120.868 \text{ amu}$$

Determine the isotopic (fractional) abundance of  $^{123}\text{Sb}$  using the average atomic mass of Sb from the periodic table.

## IN-CLASS QUESTION 2

Learning Objective(s): 2.6

There are three naturally occurring isotopes of silicon (atomic masses listed below). The percent abundance of  $^{29}\text{Si}$  is 4.68 %.

Find the percent abundance of  $^{30}\text{Si}$  using the following information and the average atomic mass of Si from the periodic table.

$$^{28}\text{Si} = 27.9769 \text{ amu}$$

$$^{29}\text{Si} = 28.9765 \text{ amu}$$

$$^{30}\text{Si} = 29.9738 \text{ amu}$$

# The Periodic Table

2.6

The periodic table organizes elements by increasing atomic number with elements having similar chemical properties aligned in vertical columns.

Groups: The vertical columns of the periodic table

- The modern periodic table labels the groups 1-18.
- Older versions use 1A-2A for the main group elements and 1B-8B for the transition metals. Roman numerals also are used frequently in older versions.

Periods: The horizontal rows of the periodic table

## ОПЫТЪ СИСТЕМЫ ЭЛЕМЕНТОВЪ.

ОСНОВАННОЙ НА ИХЪ АТОМНОМЪ ВѢСѢ И ХИМИЧЕСКОМЪ СХОДСТВѢ.

|        |          |            |              |            |             |
|--------|----------|------------|--------------|------------|-------------|
|        |          |            | Ti = 50      | Zr = 90    | ? = 180.    |
|        |          |            | V = 51       | Nb = 94    | Ta = 182.   |
|        |          |            | Cr = 52      | Mo = 96    | W = 186.    |
|        |          |            | Mn = 55      | Rh = 104,4 | Pt = 197,1. |
|        |          |            | Fe = 56      | Ru = 104,4 | Ir = 198.   |
|        |          |            | Ni = Co = 59 | Pd = 106,6 | Os = 199.   |
|        |          |            | Cu = 63,4    | Ag = 108   | Hg = 200.   |
| H = 1  | Be = 9,1 | Mg = 24    | Zn = 65,2    | Cd = 112   |             |
|        | B = 11   | Al = 27,1  | ? = 68       | Ur = 116   | Au = 197?   |
|        | C = 12   | Si = 28    | ? = 70       | Sn = 118   |             |
|        | N = 14   | P = 31     | As = 75      | Sb = 122   | Bi = 210?   |
|        | O = 16   | S = 32     | Se = 79,4    | Te = 128?  |             |
|        | F = 19   | Cl = 35,5  | Br = 80      | I = 127    |             |
| Li = 7 | Na = 23  | K = 39     | Rb = 85,4    | Cs = 133   | Tl = 204.   |
|        |          | Ca = 40    | Sr = 87,6    | Ba = 137   | Pb = 207.   |
|        |          | ? = 45     | Ce = 92      |            |             |
|        |          | ?Er = 56   | La = 94      |            |             |
|        |          | ?Yt = 60   | Di = 95      |            |             |
|        |          | ?In = 75,6 | Th = 118?    |            |             |

Д. Менделѣевъ

FIGURE 2.15 Mendeleev's Periodic Table



# PERIODIC TABLE OF ELEMENTS

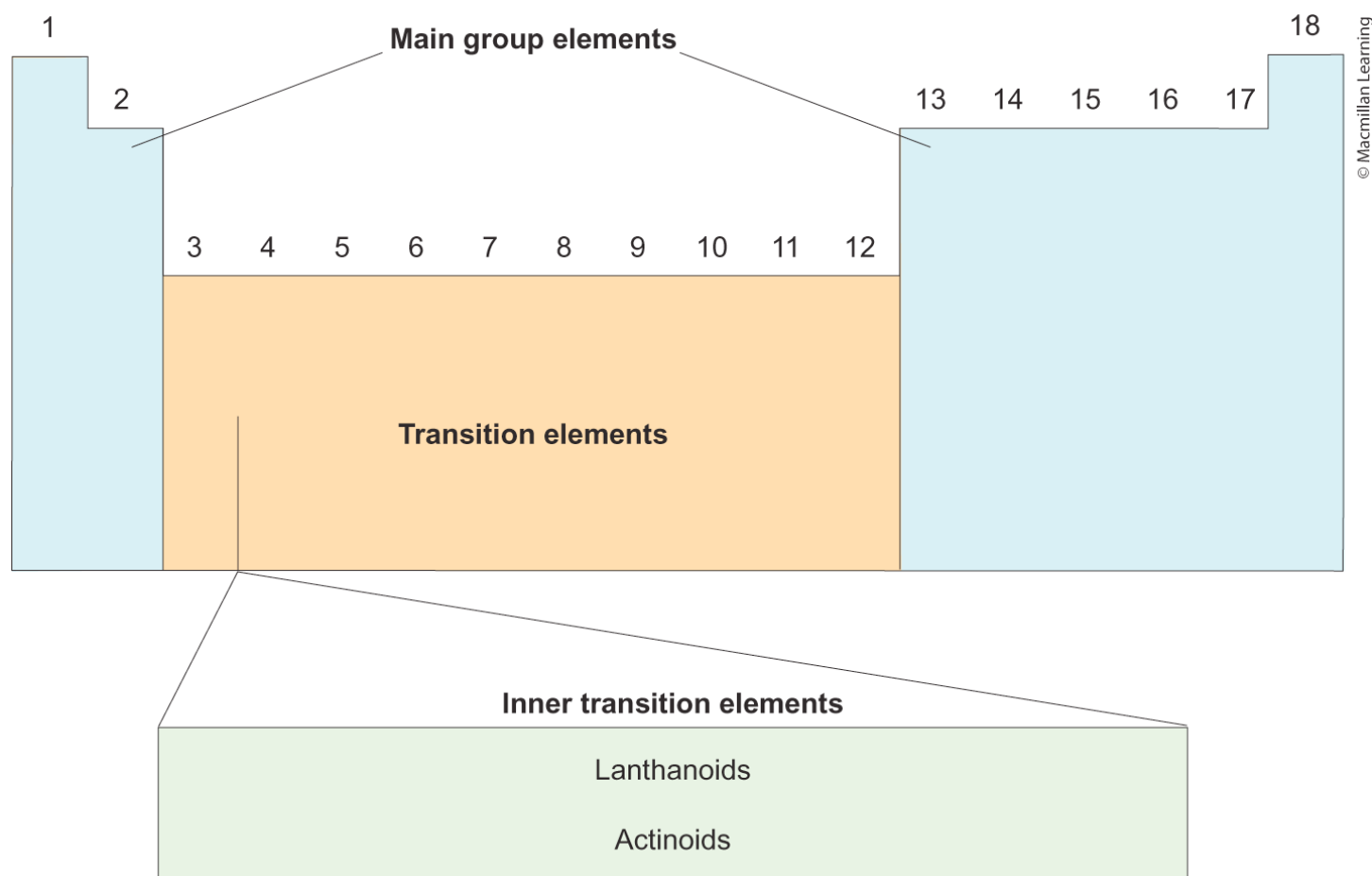
|                                                                                                                    | 1                                                                 | 2                                                                                                                           | 3                                                                 | 4                                                                      | 5                                                                 | 6                                                                     | 7                                                                  | 8                                                                  | 9                                                                   | 10                                                                    | 11                                                                   | 12                                                                   | 13                                                                  | 14                                                                  | 15                                                                 | 16                                                                   | 17                                                                  | 18                                                                  |                                                                  |                                                                |
|--------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------|------------------------------------------------------------------------|-------------------------------------------------------------------|-----------------------------------------------------------------------|--------------------------------------------------------------------|--------------------------------------------------------------------|---------------------------------------------------------------------|-----------------------------------------------------------------------|----------------------------------------------------------------------|----------------------------------------------------------------------|---------------------------------------------------------------------|---------------------------------------------------------------------|--------------------------------------------------------------------|----------------------------------------------------------------------|---------------------------------------------------------------------|---------------------------------------------------------------------|------------------------------------------------------------------|----------------------------------------------------------------|
| 1                                                                                                                  | <div>1</div> <div>H</div> <div>Hydrogen</div> <div>1.008</div>    | <div>Atomic Symbol Name Weight</div> <div><div>C Solid</div><div>Hg Liquid</div><div>H Gas</div><div>Rf Unknown</div></div> |                                                                   |                                                                        |                                                                   |                                                                       |                                                                    |                                                                    |                                                                     |                                                                       |                                                                      |                                                                      |                                                                     |                                                                     |                                                                    |                                                                      |                                                                     | <div>2</div> <div>He</div> <div>Helium</div> <div>4.0026</div>      |                                                                  |                                                                |
| 2                                                                                                                  | <div>3</div> <div>Li</div> <div>Lithium</div> <div>6.94</div>     | <div>4</div> <div>Be</div> <div>Beryllium</div> <div>9.0122</div>                                                           |                                                                   |                                                                        |                                                                   |                                                                       |                                                                    |                                                                    |                                                                     |                                                                       |                                                                      |                                                                      |                                                                     |                                                                     | <div>5</div> <div>B</div> <div>Boron</div> <div>10.81</div>        | <div>6</div> <div>C</div> <div>Carbon</div> <div>12.011</div>        | <div>7</div> <div>N</div> <div>Nitrogen</div> <div>14.007</div>     | <div>8</div> <div>O</div> <div>Oxygen</div> <div>15.999</div>       | <div>9</div> <div>F</div> <div>Fluorine</div> <div>18.998</div>  | <div>10</div> <div>Ne</div> <div>Neon</div> <div>20.180</div>  |
| 3                                                                                                                  | <div>11</div> <div>Na</div> <div>Sodium</div> <div>22.990</div>   | <div>12</div> <div>Mg</div> <div>Magnesium</div> <div>24.305</div>                                                          |                                                                   |                                                                        |                                                                   |                                                                       |                                                                    |                                                                    |                                                                     |                                                                       |                                                                      |                                                                      |                                                                     |                                                                     | <div>13</div> <div>Al</div> <div>Aluminium</div> <div>26.982</div> | <div>14</div> <div>Si</div> <div>Silicon</div> <div>28.085</div>     | <div>15</div> <div>P</div> <div>Phosphorus</div> <div>30.974</div>  | <div>16</div> <div>S</div> <div>Sulfur</div> <div>32.06</div>       | <div>17</div> <div>Cl</div> <div>Chlorine</div> <div>35.45</div> | <div>18</div> <div>Ar</div> <div>Argon</div> <div>39.948</div> |
| 4                                                                                                                  | <div>19</div> <div>K</div> <div>Potassium</div> <div>39.098</div> | <div>20</div> <div>Ca</div> <div>Calcium</div> <div>40.078</div>                                                            | <div>21</div> <div>Sc</div> <div>Scandium</div> <div>44.956</div> | <div>22</div> <div>Ti</div> <div>Titanium</div> <div>47.867</div>      | <div>23</div> <div>V</div> <div>Vanadium</div> <div>50.942</div>  | <div>24</div> <div>Cr</div> <div>Chromium</div> <div>51.996</div>     | <div>25</div> <div>Mn</div> <div>Manganese</div> <div>54.938</div> | <div>26</div> <div>Fe</div> <div>Iron</div> <div>55.845</div>      | <div>27</div> <div>Co</div> <div>Cobalt</div> <div>58.933</div>     | <div>28</div> <div>Ni</div> <div>Nickel</div> <div>58.693</div>       | <div>29</div> <div>Cu</div> <div>Copper</div> <div>63.546</div>      | <div>30</div> <div>Zn</div> <div>Zinc</div> <div>65.38</div>         | <div>31</div> <div>Ga</div> <div>Gallium</div> <div>69.723</div>    | <div>32</div> <div>Ge</div> <div>Germanium</div> <div>72.630</div>  | <div>33</div> <div>As</div> <div>Arsenic</div> <div>74.922</div>   | <div>34</div> <div>Se</div> <div>Selenium</div> <div>78.971</div>    | <div>35</div> <div>Br</div> <div>Bromine</div> <div>79.904</div>    | <div>36</div> <div>Kr</div> <div>Krypton</div> <div>83.798</div>    |                                                                  |                                                                |
| 5                                                                                                                  | <div>37</div> <div>Rb</div> <div>Rubidium</div> <div>85.468</div> | <div>38</div> <div>Sr</div> <div>Strontium</div> <div>87.62</div>                                                           | <div>39</div> <div>Y</div> <div>Yttrium</div> <div>88.906</div>   | <div>40</div> <div>Zr</div> <div>Zirconium</div> <div>91.224</div>     | <div>41</div> <div>Nb</div> <div>Niobium</div> <div>92.906</div>  | <div>42</div> <div>Mo</div> <div>Molybdenum</div> <div>95.95</div>    | <div>43</div> <div>Tc</div> <div>Technetium</div> <div>(98)</div>  | <div>44</div> <div>Ru</div> <div>Ruthenium</div> <div>101.07</div> | <div>45</div> <div>Rh</div> <div>Rhodium</div> <div>102.91</div>    | <div>46</div> <div>Pd</div> <div>Palladium</div> <div>106.42</div>    | <div>47</div> <div>Ag</div> <div>Silver</div> <div>107.87</div>      | <div>48</div> <div>Cd</div> <div>Cadmium</div> <div>112.41</div>     | <div>49</div> <div>In</div> <div>Indium</div> <div>114.82</div>     | <div>50</div> <div>Sn</div> <div>Tin</div> <div>118.71</div>        | <div>51</div> <div>Sb</div> <div>Antimony</div> <div>121.76</div>  | <div>52</div> <div>Te</div> <div>Tellurium</div> <div>127.60</div>   | <div>53</div> <div>I</div> <div>Iodine</div> <div>126.90</div>      | <div>54</div> <div>Xe</div> <div>Xenon</div> <div>131.29</div>      |                                                                  |                                                                |
| 6                                                                                                                  | <div>55</div> <div>Cs</div> <div>Caesium</div> <div>132.91</div>  | <div>56</div> <div>Ba</div> <div>Barium</div> <div>137.33</div>                                                             | <div>57–71</div> <div>89–103</div>                                | <div>72</div> <div>Hf</div> <div>Hafnium</div> <div>178.49</div>       | <div>73</div> <div>Ta</div> <div>Tantalum</div> <div>180.95</div> | <div>74</div> <div>W</div> <div>Tungsten</div> <div>183.84</div>      | <div>75</div> <div>Re</div> <div>Rhenium</div> <div>186.21</div>   | <div>76</div> <div>Os</div> <div>Osmium</div> <div>190.23</div>    | <div>77</div> <div>Ir</div> <div>Iridium</div> <div>192.22</div>    | <div>78</div> <div>Pt</div> <div>Platinum</div> <div>195.08</div>     | <div>79</div> <div>Au</div> <div>Gold</div> <div>196.97</div>        | <div>80</div> <div>Hg</div> <div>Mercury</div> <div>200.59</div>     | <div>81</div> <div>Tl</div> <div>Thallium</div> <div>204.38</div>   | <div>82</div> <div>Pb</div> <div>Lead</div> <div>207.2</div>        | <div>83</div> <div>Bi</div> <div>Bismuth</div> <div>208.98</div>   | <div>84</div> <div>Po</div> <div>Polonium</div> <div>(209)</div>     | <div>85</div> <div>At</div> <div>Astatine</div> <div>(210)</div>    | <div>86</div> <div>Rn</div> <div>Radon</div> <div>(222)</div>       |                                                                  |                                                                |
| 7                                                                                                                  | <div>87</div> <div>Fr</div> <div>Francium</div> <div>(223)</div>  | <div>88</div> <div>Ra</div> <div>Radium</div> <div>(226)</div>                                                              |                                                                   | <div>104</div> <div>Rf</div> <div>Rutherfordium</div> <div>(267)</div> | <div>105</div> <div>Db</div> <div>Dubnium</div> <div>(268)</div>  | <div>106</div> <div>Sg</div> <div>Seaborgium</div> <div>(269)</div>   | <div>107</div> <div>Bh</div> <div>Bohrium</div> <div>(270)</div>   | <div>108</div> <div>Hs</div> <div>Hassium</div> <div>(277)</div>   | <div>109</div> <div>Mt</div> <div>Meitnerium</div> <div>(278)</div> | <div>110</div> <div>Ds</div> <div>Darmstadtium</div> <div>(281)</div> | <div>111</div> <div>Rg</div> <div>Roentgenium</div> <div>(282)</div> | <div>112</div> <div>Cn</div> <div>Copernicium</div> <div>(285)</div> | <div>113</div> <div>Nh</div> <div>Nihonium</div> <div>(286)</div>   | <div>114</div> <div>Fl</div> <div>Flerovium</div> <div>(289)</div>  | <div>115</div> <div>Mc</div> <div>Moscovium</div> <div>(290)</div> | <div>116</div> <div>Lv</div> <div>Livermorium</div> <div>(293)</div> | <div>117</div> <div>Ts</div> <div>Tennessine</div> <div>(294)</div> | <div>118</div> <div>Og</div> <div>Oganesson</div> <div>(294)</div>  |                                                                  |                                                                |
| For elements with no stable isotopes, the mass number of the isotope with the longest half-life is in parentheses. |                                                                   |                                                                                                                             |                                                                   |                                                                        |                                                                   |                                                                       |                                                                    |                                                                    |                                                                     |                                                                       |                                                                      |                                                                      |                                                                     |                                                                     |                                                                    |                                                                      |                                                                     |                                                                     |                                                                  |                                                                |
|                                                                                                                    |                                                                   |                                                                                                                             | 6                                                                 | <div>57</div> <div>La</div> <div>Lanthanum</div> <div>138.91</div>     | <div>58</div> <div>Ce</div> <div>Cerium</div> <div>140.12</div>   | <div>59</div> <div>Pr</div> <div>Praseodymium</div> <div>140.91</div> | <div>60</div> <div>Nd</div> <div>Neodymium</div> <div>144.24</div> | <div>61</div> <div>Pm</div> <div>Promethium</div> <div>(145)</div> | <div>62</div> <div>Sm</div> <div>Samarium</div> <div>150.36</div>   | <div>63</div> <div>Eu</div> <div>Europium</div> <div>151.96</div>     | <div>64</div> <div>Gd</div> <div>Gadolinium</div> <div>157.25</div>  | <div>65</div> <div>Tb</div> <div>Terbium</div> <div>158.93</div>     | <div>66</div> <div>Dy</div> <div>Dysprosium</div> <div>162.50</div> | <div>67</div> <div>Ho</div> <div>Holmium</div> <div>164.93</div>    | <div>68</div> <div>Er</div> <div>Erbium</div> <div>167.26</div>    | <div>69</div> <div>Tm</div> <div>Thulium</div> <div>168.93</div>     | <div>70</div> <div>Yb</div> <div>Ytterbium</div> <div>173.05</div>  | <div>71</div> <div>Lu</div> <div>Lutetium</div> <div>174.97</div>   |                                                                  |                                                                |
|                                                                                                                    |                                                                   |                                                                                                                             | 7                                                                 | <div>89</div> <div>Ac</div> <div>Actinium</div> <div>(227)</div>       | <div>90</div> <div>Th</div> <div>Thorium</div> <div>232.04</div>  | <div>91</div> <div>Pa</div> <div>Protactinium</div> <div>231.04</div> | <div>92</div> <div>U</div> <div>Uranium</div> <div>238.03</div>    | <div>93</div> <div>Np</div> <div>Neptunium</div> <div>(237)</div>  | <div>94</div> <div>Pu</div> <div>Plutonium</div> <div>(244)</div>   | <div>95</div> <div>Am</div> <div>Americium</div> <div>(243)</div>     | <div>96</div> <div>Cm</div> <div>Curium</div> <div>(247)</div>       | <div>97</div> <div>Bk</div> <div>Berkelium</div> <div>(247)</div>    | <div>98</div> <div>Cf</div> <div>Californium</div> <div>(251)</div> | <div>99</div> <div>Es</div> <div>Einsteinium</div> <div>(252)</div> | <div>100</div> <div>Fm</div> <div>Fermium</div> <div>(257)</div>   | <div>101</div> <div>Md</div> <div>Mendelevium</div> <div>(258)</div> | <div>102</div> <div>No</div> <div>Nobelium</div> <div>(259)</div>   | <div>103</div> <div>Lr</div> <div>Lawrencium</div> <div>(266)</div> |                                                                  |                                                                |

Ptable

# The Periodic Table

2.6

The elements also can be grouped in larger categories:



**FIGURE 2.20**

# The Periodic Table

2.6

The elements also can be grouped in larger categories:

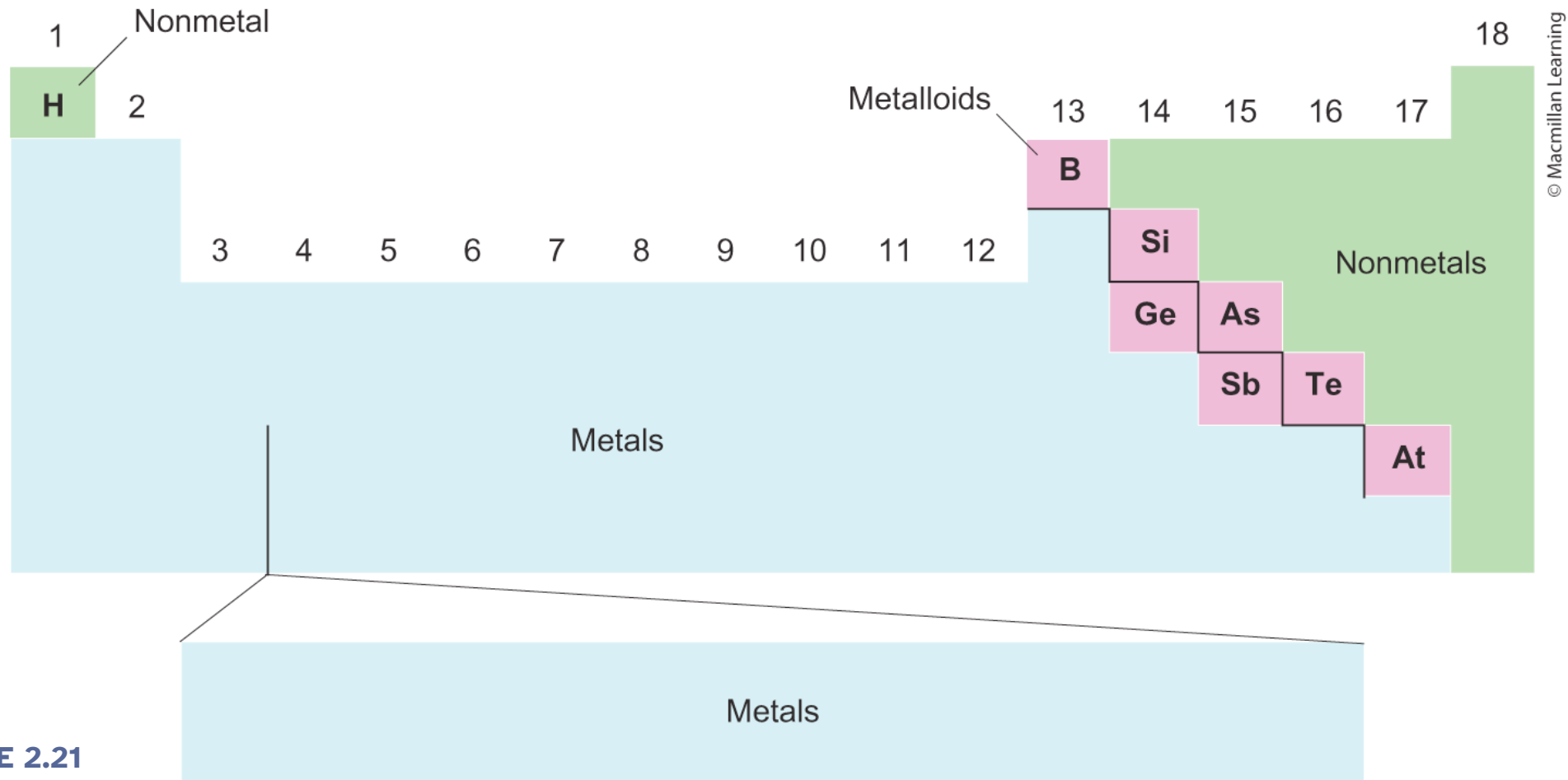


FIGURE 2.21

## IN-CLASS QUESTION 3

Learning Objective(s): 2.7, 2.8

Which of the following are classified as nonmetals?

☐ Si

☐ Ne

☐ Fe

☐ Sr

☐ Br

## IN-CLASS QUESTION 4

Learning Objective(s): 2.7, 2.8

What is the chemical symbol of the alkali metal in the fourth period?

# Terminology

## Chemistry in Context

It's important to understand chemical vocabulary.

Atom: Smallest part of an element that is still that element.

Molecule: Two or more atoms joined together and acting as a unit.

Compound: Two or more *different* elements joined together.

Chemists/Scientists use the term element in different ways though.



# Chemical Formulas

3.1

Atomic elements: Most elements exist in nature as individual neutral atoms.

Molecular elements: A few elements exist in nature as small molecules containing two or more atoms bonded together.



(a)



(b)



(c)

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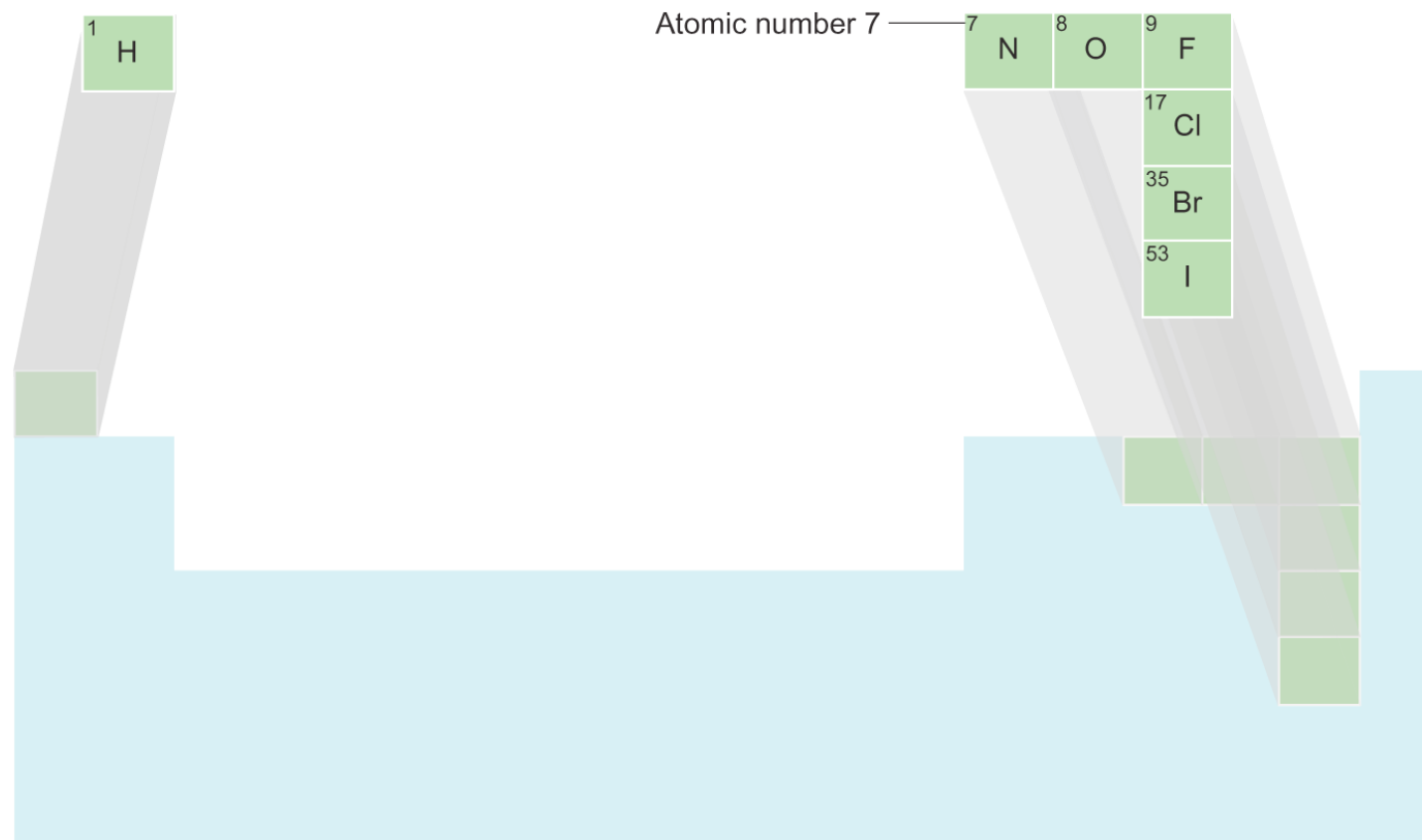
**FIGURE 3.2** Examples of molecular elements (a)  $\text{Cl}_2$ , (b)  $\text{P}_4$ , (c)  $\text{S}_8$ .

# Chemical Formulas

## 3.1

**Molecular diatomic elements:** Seven elements occur as diatomic molecules when they are in their elemental/natural form.

- Hydrogen
- Nitrogen
- Oxygen
- Fluorine
- Chlorine
- Bromine
- Iodine



**FIGURE 3.3 The Seven Elements that Exist as Diatomic Molecules**

# Chemical Formulas

3.1

Chemical formulas combine elemental symbols and subscripts to represent chemical compounds.

- Provide information about the relative number of atoms of each element
- Sometimes also give information about the arrangement of atoms in the molecule.

The group of atoms represented by a chemical formula is called a formula unit.

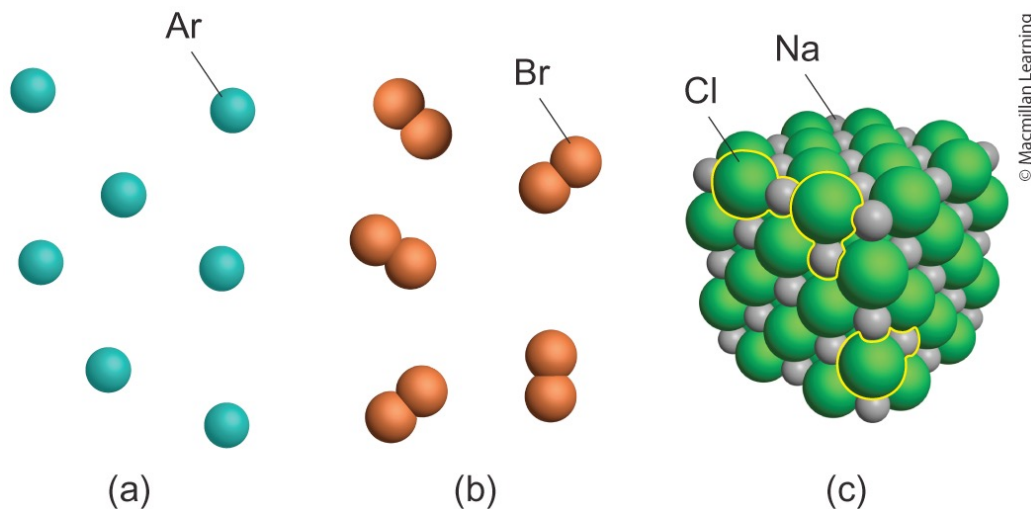
- An atom or molecule of an uncombined element, a molecule of a covalent compound, or a set of ions in an ionic compound

# Chemical Formulas

## 3.1

Compounds can exist in different forms.

- Molecular or covalent compounds: Exist as individual particles (called molecules) made up of bonded nonmetal elements.
- Ionic compounds: Exist not as individual particles but as extended, 3D lattice structures. The simplest repeating unit is called a formula unit.



**FIGURE 3.1** (a) atoms of Ar,  
(b) molecules of  $\text{Br}_2$ ,  
(c) lattice of NaCl

# Chemical Formulas

3.1

Chemists combine chemical symbols with subscripted numbers to form chemical formulas.

The subscripted numbers indicate the number of atoms of each element present.

